CLASS 9 — MATHS · NCERT-TYPE CHAPTER TEST SET 3

Chapter 2: **Linear Polynomials** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The graph of $y = 5x - 25$ cuts the x-axis at: 1

(a) $(5, 0)$ **(b)** $(6, 0)$ **(c)** $(-5, 0)$ **(d)** $(0, 5)$

2. The sum of the zeros of $x - 7$ and $6x - 18$ is: 1

(a) 9 **(b)** 10 **(c)** 11 **(d)** 12

3. The nth term of the pattern 3, 12, 21, 30, ... is: 1

(a) $9n - 6$ **(b)** $9n - 7$ **(c)** $9n - 5$ **(d)** $10n - 6$

4. A plumber charges ₹80 plus ₹40 per hour. For 5 hours he charges ₹: 1

(a) 281 **(b)** 280 **(c)** 282 **(d)** 279

5. Which of these is a linear polynomial? 1

(a) $4x^3$ **(b)** $4x$ **(c)** $4x^2$ **(d)** 4

6. If $x = 6$ is a zero of the polynomial $4x + k$, then $k =$ 1

(a) -22 **(b)** -24 **(c)** -25 **(d)** -23

7. The value of $4x + 9$ at $x = -5$ is: 1

(a) -11 **(b)** -10 **(c)** -12 **(d)** -9

8. The line $y = -2x + 9$ passes through $(2, p)$. The value of p is: 1

(a) 4 **(b)** 7 **(c)** 5 **(d)** 6

9. The graph of $y = 4x + 16$ cuts the x-axis at: 1

(a) $(4, 0)$ **(b)** $(-3, 0)$ **(c)** $(0, -4)$ **(d)** $(-4, 0)$

10. The sum of the zeros of $5x - 25$ and $2x - 16$ is: 1

(a) 15 **(b)** 13 **(c)** 12 **(d)** 14

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Write the slope and intercept of $y = 9x - 2$, and the coordinates of the point where it cuts the y-axis. 2

12. Find the zero of $7x - 28$ and show that it lies on the x-axis of the graph $y = 7x - 28$. 2

13. Find $p(3)$ and $p(-3)$ for $p(x) = 5x + 7$. 2

14. Write the slope and intercept of $y = -7x - 4$, and the coordinates of the point where it cuts the y-axis. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Draw up a table for $y = 5x - 3$ for $x = -1, 0, 1, 2$ and state the slope and the y-intercept. 3

16. The line $y = 2x + k$ has slope 2 and passes through $(5, 3)$. Find k and write the line. 3

17. If $p(x) = 3x - 6$ and $q(x) = 3x$, find $p(x) + q(x)$ and its zero. 3

OR

Draw up a table for $y = -3x + 6$ for $x = -1, 0, 1, 2$ and state the slope and the y-intercept. 3

18. The line $y = 3x + k$ has slope 3 and passes through $(4, 20)$. Find k and write the line. 3

OR

If $p(x) = 2x - 10$ and $q(x) = 3x$, find $p(x) + q(x)$ and its zero. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. The cost of x notebooks is a linear polynomial: a shop charges ₹55 for the bag plus ₹11 per notebook. Write the cost $C(x)$, find the cost of 7 notebooks and how many notebooks can be bought for ₹121. 5

OR

For $y = 6x - 18$: 5
(i) find y at $x = 0, 1, 2$;
(ii) find the point where the graph crosses the x-axis;
(iii) state the slope and say whether the graph rises or falls from left to right.

20. The cost of x notebooks is a linear polynomial: a shop charges ₹120 for the bag plus ₹25 per notebook. Write the cost $C(x)$, find the cost of 2 notebooks and how many notebooks can be bought for ₹395. 5

OR

For $y = -4x + 20$: 5
(i) find y at $x = 0, 1, 2$;
(ii) find the point where the graph crosses the x-axis;
(iii) state the slope and say whether the graph rises or falls from left to right.

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND